Article

On the minimum dataset size and quality for a plant counting object detector

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Abstract: A single paragraph of about 200 words maximum. For research articles, abstracts should give a pertinent overview of the work. We strongly encourage authors to use the following style of structured abstracts, but without headings: (1) Background: place the question addressed in a broad context and highlight the purpose of the study; (2) Methods: describe briefly the main methods or treatments applied; (3) Results: summarize the article's main findings; (4) Conclusions: indicate the main conclusions or interpretations. The abstract should be an objective representation of the article, it must not contain results which are not presented and substantiated in the main text and should not exaggerate the main conclusions.

Keywords: keyword 1; keyword 2; keyword 3 (List three to ten pertinent keywords specific to the article; yet reasonably common within the subject discipline.)

1. Introduction

Plant counting is a critical operation in precision agriculture, plant breeding, and agronomical managing evaluation. Accurate plant counts can provide valuable information for both farmers and researchers. This task was often performed manually by human operators. Now a days this process can be automated by the use of computer vision algorithms. To validate a method for counting, it is critical to set a benchmark for accuracy. A benchamrk can be defined by the accuracy of manual counting, international standards, or by comparison with other methods. Accuracy of plant manual counting is difficult to determine, because it depends on many factors and no clear results are shown in litterature, but it is often taken as golden sample. According to the European Plant Protection Organization, the benchmark for acceptance is a coefficient of determination (r squared) of 0.85 when compared to manual counting [1]. This corresponds to a Root Mean Square Error (RMSE) of approximately 0.39. Also scientific literature mention a r-squared value of 0.85 with ground truth (manual counting) as a benchmark for acceptance [2]. Litterature shows that benchmark for acceptance can be achieved with object detection [3–5] or regression models [6,7]. A superiority of object detection over regression models in terms of accuracy was found [2]. Object detection is also more versatile than regression models, because, inference on georeferenced orthomosaics can be used to verify plant presence through its coordinates throughout the vegetative phases.

Object detection algorithms are designed to identify and locate objects within an image. Identification supports are usually bounding boxes, which are rectangles that enclose the

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object of interest, but they can also be points or other kind of polygons. Counting of plants is then commonly performed by counting the number of bounding boxes that enclose the objects of interest (the plant) in a image area. Georeferenced orthomosaics are images created through aerial photogrammetry, a process that involves capturing overlapping aerial georeferenced images taken in nadiral view and georeferenced ground control points, then performing boundle adjustment to form a single, seamless image [8]. Georeferenced orthomosaics are scaled and oriented to a geographical coordinate system. It implies that the pixel coordinates and size correspond to geographical coordinates and are projectables in a metric system. Using georeferenced orthomosaics in seedling counting makes object detection easier, because the metric scale can be used to locate the objects in the image and prospective variance is reduced.

The development of object detection algorithms has evolved from not machine learning methods, here named handcrafted methods (HC), to modern deep learning-based (DL) techniques. Pratically all the state-of-the-art object detection methods are based on DL models now a days. Nethertheless, HC methods are still used in some cases, in particular to extract the in-domain dataset for training a DL model. Modern DL object detection uses CNN-based frameworks (e.g., Faster R-CNN [9], YOLO [10]) and Transformer-based approaches (e.g., DETR [11]). The main difference between CNN-based and Transformer-based models is the way they process the image. CNN-based models process the image in a grid-like fashion, while Transformer-based models process the image as a sequence of patches. On common benchmarks as COCO, PASCAL VOC, and ImageNet, Transformer-based models have shown to be more accurate than CNN-based models. Nethertheless CNN-based models are still widely used in object detection, because they are more efficient in processing small images and have a lower computational cost. One of the critical challenges in training effective DL object detection models is the availability of a sufficiently large and high-quality dataset. Despite the superiority of DL models on HC methods, it is proven that HC can be used to get an initial dataset to train a DL model, effectivelly reducing the annotation burden, and in some cases automating it. Recently, zero-shot and few-shot object detection have emerged as promising paradigms to alleviate the need for large annotated datasets. While traditional object detection models require extensive labeled data for training, few-shot and zero-shot object detection aim to detect novel objects with little to no labeled examples respectivelly. They often leverage feature transfer or meta-learning components to generalize across classes under extreme data scarcity. This approach reduces the annotation burden and is especially beneficial in domains where collecting exhaustive training data is impractical. Zero-shot object detection detects new categories without any training samples by leveraging semantic relationships or contextual information learned from known classes. Few-shot approaches optimize models to learn quickly from a handful of labeled examples. Zero-shot actual state-of-the-art models are YOLO-World, OWLv2, and Grounding DINO. DE-ViT and CD-ViTO are the latest models that have shown promising results in few-shot object detection.

Many studies have been conducted on object detection for plants in open field [3,5–7,12–21], but few have focused on minimum requirements for training a robust object detector [4,22], and only one specifically tried to minimize the size of the training dataset to 600 512*512 RGB images [23]. Dataset size and quality predictability in DL is a controversial topic that has been proven to be addressable with empirical approaches [24,25]. It is well known that the performance of a DL model is directly related to the amount of data used for training [26], its quality [27]. The model architecture is critical in dataset requirements as different models may require different dataset sizes and qualities to achieve the same performance [28,29]. Backbone is pivotal on downstream tasks as object detection. The importance of using a domain specific backbone has been proven also for plant/leaves segmentation on orthomosaics [30], but backbone training is still proibitive for many applications.

Nethertheless, the most important factor affecting the dataset size for a plant counting object detector is the dataset source. It has been observed that using training samples from the same dataset as the inference (in-domain dataset) increases accuracy and reduce the need for a large dataset in respect to collecting training samples from other datasets (out-of-distribution dataset) [4,22].

Plant counting is a challenging task, mostly because of data scarcity: public datasets are rare and often not suitable for the task because of the large environmental variability, their images are not orthorectified or scale is unknown. Even so we can find some datasets that can be used for training a plant counting object detector. We can reduce the variability that the model should learn by constrain the dataset to a specific crop at a specific growth stage. For this study we choose maize seedlings (Zea mais L.) at the V3-V5 (BBCH 13-15) growth stage. It is a common crop that in this particular growth stage is easy to count because of low overlapping and fixed intra-row and inter-row spacing.

This study aims to determine the minimum dataset size and quality required to train a benchmark reaching object detection model for identifying maize seedlings in georeferenced orthomosaics, where the dataset size and quality are respectivelly defined by the amount of annotated images in the training set, and the accuracy of the annotations. Also the effect of training dataset source will be evaluated in this study. Models architecture and size will be taken into account, as long as the object detection downstream task is concerned. After setting the dataset size and quality minimum requirements for traditional DL object detectors, we will evaluate if new zero-shot and few-shot object detection models can achieve the same benchmark for acceptance with less data and annotations.

2. Materials and Methods

Materials and Methods should be described with sufficient details to allow others to replicate and build on published results. Please note that publication of your manuscript implicates that you must make all materials, data, computer code, and protocols associated with the publication available to readers. Please disclose at the submission stage any restrictions on the availability of materials or information. New methods and protocols should be described in detail while well-established methods can be briefly described and appropriately cited.

Research manuscripts reporting large datasets that are deposited in a publicly available database should specify where the data have been deposited and provide the relevant accession numbers. If the accession numbers have not yet been obtained at the time of submission, please state that they will be provided during review. They must be provided prior to publication.

Interventionary studies involving animals or humans, and other studies require ethical approval must list the authority that provided approval and the corresponding ethical approval code.

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3. Results

This section may be divided by subheadings. It should provide a concise and precise description of the experimental results, their interpretation as well as the experimental conclusions that can be drawn.

3.1. Subsection

3.1.1. Subsubsection

Bulleted lists look like this:

- First bullet;
- Second bullet;
- Third bullet.

Numbered lists can be added as follows:

1. First item;
2. Second item;
3. Third item.

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All figures and tables should be cited in the main text as Figure 1, Table 1, etc.

Figure 1. This is a figure. Schemes follow the same formatting.

Table 1. This is a table caption. Tables should be placed in the main text near to the first time they are cited.

Title 1	**Title 2**	**Title 3**
Entry 1	Data	Data
Entry 2	Data	Data [1]

[1] Tables may have a footer.

The text continues here (Figure 2 and Table 2).

(**a**) (**b**)

(**c**) (**d**)

Figure 2. This is a wide figure. Schemes follow the same formatting. If there are multiple panels, they should be listed as: (**a**) Description of what is contained in the first panel. (**b**) Description of what is contained in the second panel. (**c**) Description of what is contained in the third panel. (**d**) Description of what is contained in the fourth panel. Figures should be placed in the main text near to the first time they are cited. A caption on a single line should be centered.

Table 2. This is a wide table.

Title 1	**Title 2**	**Title 3**	**Title 4**
Entry 1 *	Data	Data	Data
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	Data	Data	Data
Entry 2	Data	Data	Data
	Data	Data	Data
	Data	Data	Data

* Tables may have a footer.

Text.
Text.

3.3. Formatting of Mathematical Components

This is the example 1 of equation:

$$a = 1, \tag{1}$$

the text following an equation need not be a new paragraph. Please punctuate equations as regular text.

This is the example 2 of equation:

$$a = b + c + d + e + f + g + h + i + j + k + l + m + n + o + p + q + r + s + t + u + v + w + x + y + z \tag{2}$$

Table 3. This is a very wide table.

Title 1	**Title 2**	**Title 3**	**Title 4**
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Entry 2	Data	Data	Data[1]

[1] This is a table footnote.

Please punctuate equations as regular text. Theorem-type environments (including propositions, lemmas, corollaries etc.) can be formatted as follows:

Theorem 1. *Example text of a theorem.*

The text continues here. Proofs must be formatted as follows:

Proof of Theorem 1. Text of the proof. Note that the phrase "of Theorem 1" is optional if it is clear which theorem is being referred to. □

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4. Discussion

Authors should discuss the results and how they can be interpreted from the perspective of previous studies and of the working hypotheses. The findings and their implications should be discussed in the broadest context possible. Future research directions may also be highlighted.

5. Conclusions

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Abbreviations

The following abbreviations are used in this manuscript:

MDPI	Multidisciplinary Digital Publishing Institute
DOAJ	Directory of open access journals
TLA	Three letter acronym
LD	Linear dichroism

Appendix A

Appendix A.1

The appendix is an optional section that can contain details and data supplemental to the main text—for example, explanations of experimental details that would disrupt the flow of the main text but nonetheless remain crucial to understanding and reproducing the research shown; figures of replicates for experiments of which representative data are shown in the main text can be added here if brief, or as Supplementary Data. Mathematical proofs of results not central to the paper can be added as an appendix.

Table A1. This is a table caption.

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Entry 2	Data	Data

Appendix B

All appendix sections must be cited in the main text. In the appendices, Figures, Tables, etc. should be labeled, starting with "A"—e.g., Figure A1, Figure A2, etc.

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